

Project Overview

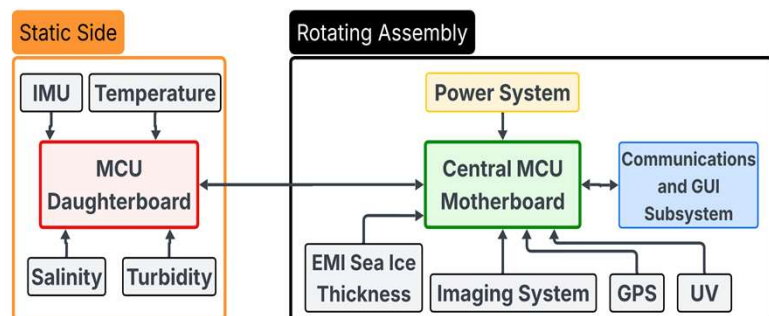
Problem: The Arctic is warming four times faster than the global average, yet continuous measurement of sea-ice thickness at scale remains a challenge. Current methods are destructive or rely on human operation, creating a data gap in polar climate research.

Aim: To develop an autonomous, scalable buoy to non-destructively measure sea-ice thickness and other key environmental variables continuously across multiple seasons.

Objectives:

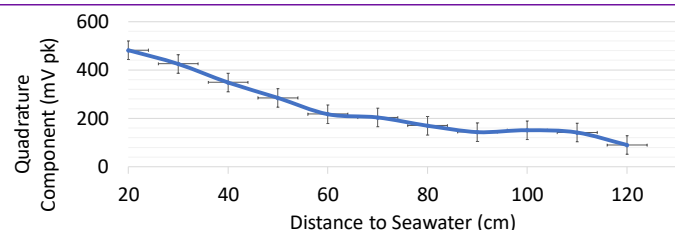
- **Advanced Sensing:** Monitoring sea-ice thickness, a range of essential climate variables and an imaging system.
- **Autonomous Operation:** A solar-charging battery system engineered to survive months of polar winter darkness.
- **Air-Deployable:** A compact, self-levelling design built for widespread mass deployment.
- **Satellite Communication:** Wireless data retrieval to user interface.

System Architecture

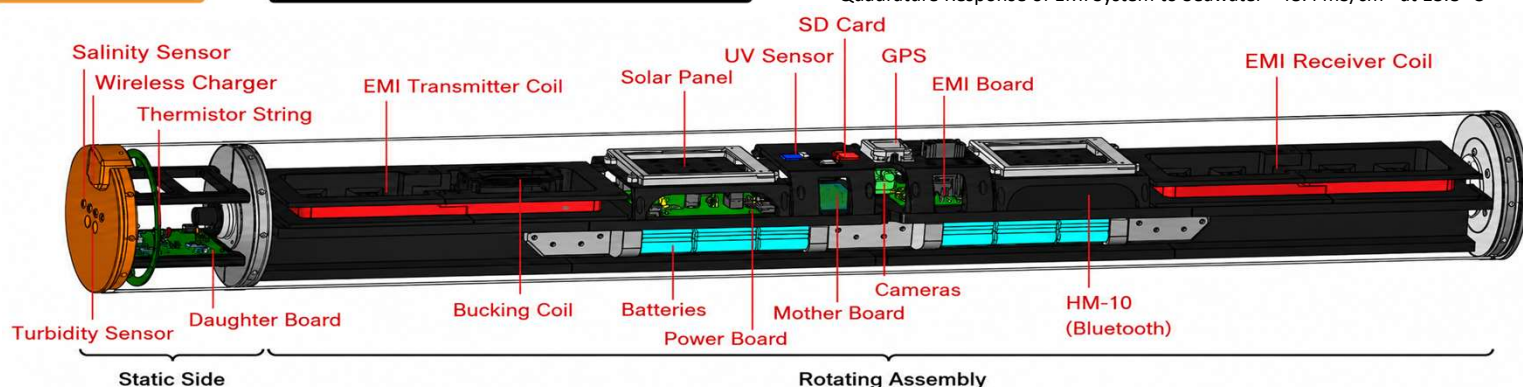


EMI Sea Ice Thickness Measurement

- A custom electromagnetic induction (EMI) system non-destructively measures sea-ice thickness; optimising coil layout within sonobuoy.

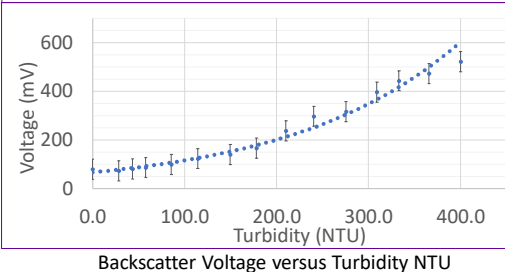


Quadrature Response of EMI System to Seawater - 48.4 mS/cm³ at 13.8 °C

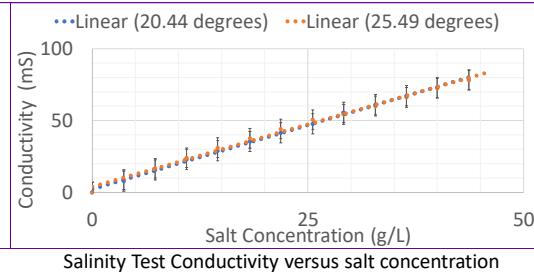


Environmental Sensors

- **Turbidity:** Monitors water clarity to detect sediment and biological changes below the ice.
- **Salinity:** Custom 4 probe conductivity sensor that tracks salt concentration changes.
- **Temperature Probes:** Records external ice and sea water temperatures irrespective of the buoy's orientation using a thermistor string.
- **Cameras:** Captures visual records of the local environment, with onboard Otsu-based filtering to reject poor-quality images and minimise transmission bandwidth.



Backscatter Voltage versus Turbidity NTU



Salinity Test Conductivity versus salt concentration

Power System

- Battery system capable of powering buoy through winter.
- Solar panels recharge the batteries when sunlight returns, for long term operation.

Achievements

EMI Sea-Ice	1.38 m target detection range
Power Architecture	105-day winter autonomy at -40 °C solar harvesting.
Environmental Suite	Turbidity, salinity, temperature, UV, and wave sensors rated down to -40 °C.
Mechanical Integration	Complete electronic consolidation within a NATO Class B self-levelling chassis.
Telemetry	Comms link enabling 12-channel live data update

Conclusion

The project has successfully delivered a fully integrated, autonomous polar monitoring buoy capable of non-destructive sea-ice thickness measurement alongside validated environmental sensors.

Future Work

- Use satellite telemetry to replace the BLE communication.
- Develop air deployment mechanism and decide chassis materials.
- Conduct Arctic field deployment to validate real-world performance.